

MSE 360: Kinetic Processes in Materials

Homework #1

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

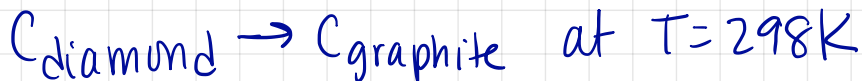
1. Define kinetics. Contrast kinetics and thermodynamics.

Kinetics deals with rates of processes (reactions + mass transport); and what factors affect these rates.

thermo: what should happen

kinetics: how fast it will happen

2. Give an example of a kinetically frustrated process that is thermodynamically favorable.



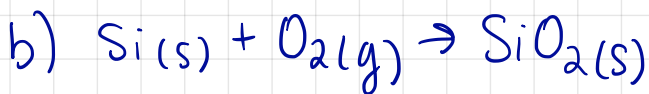
$$\Delta G_{\text{rxn}}^{\circ} = -165 \text{ J/mol}$$

But it is kinetically frustrated.

3. (a) Give the definition of a heterogeneous kinetic process.

(b) Give an example of a heterogeneous kinetic process. Provide diagrams, text, and equations to fully support your example in sufficient detail.

a) heterogeneous kinetics: involve more than 1 phase



4. The SI value of the gas constant R is $8.314 \text{ J}/(\text{mol} \cdot \text{K})$. Convert R to the following units and show all your work in each case:

- a. $\text{cal}/(\text{mol} \cdot \text{K})$
- b. $\text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$
- c. $\text{cm}^3 \cdot \text{atm}/(\text{mol} \cdot \text{K})$

a) $\text{cal}/\text{mol} \cdot \text{K}$

$$8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \times \frac{1 \text{ cal}}{4.184 \text{ J}} = \boxed{1.987 \text{ cal}/\text{mol} \cdot \text{K}}$$

b) $\text{L} \cdot \text{atm}/\text{mol} \cdot \text{K}$

$$\frac{8.314 \text{ J}}{\text{mol} \cdot \text{K}} \times \frac{1 \text{ N} \cdot \text{m}}{1 \text{ J}} \times \frac{1 \text{ Pa} \cdot \text{m}^2}{1 \text{ N}} \times \frac{1 \text{ atm}}{101,325 \text{ Pa}} \times \frac{1 \text{ L}}{10^{-3} \text{ m}^3}$$

$$\boxed{R = 0.082 \text{ L} \cdot \text{atm}/\text{mol} \cdot \text{K}}$$

c) $\text{cm}^3 \cdot \text{atm}/\text{mol} \cdot \text{K}$

$$\frac{0.082 \text{ L} \cdot \text{atm}}{\text{mol} \cdot \text{K}} \times \frac{10^{-3} \text{ m}^3}{1 \text{ L}} \times \frac{(100 \text{ cm})^3}{1 \text{ m}^3}$$

$$\boxed{R = 82 \text{ cm}^3 \cdot \text{atm}/\text{mol} \cdot \text{K}}$$